

Integrated Lean Logistics - Warehousing model to reduce Lead Time in an SME of food sector: A research in Peru

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ABSTRACT

The food industry represented 2019 3.2% of the Gross Domestic Product (GDP) in Peru, in addition to presenting high employability being that the sector contained 7.4% of the Economically Active Population (EAP) [1]. However, due to the world crisis caused by COVID 19, this sector has been harmed in the year 2020 because the GDP of the accommodation and restaurants sector decreased by 50.2% compared to the previous year [2]. For that reason, strengthening this industry is considered crucial. One of the most critical links in the management of the supply chain of material, this problem results in the late delivery of orders to their customers, which would generate significant monetary losses and a bad reputation. In this sense, the present research proposes an integral improvement model based on the combination of tools such as the 5S ordering of raw material under a multi-criteria ABC approach, FEFO, MRP, Forecasting, and BPM, whose objective is to reduce the lead time. The model was validated through a simulation in Arena 16.1, where a reduction in delivery time and the frequency of extra purchases in 7.2% y 50%.

CCS CONCEPTS

• **Applied computing**; • **Transportation**; • **Supply Chain Management**;

KEYWORDS

Foodservice, SMEs, Order fulfillment

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1 INTRODUCTION

Internationally, the hotel and restaurant sector was affected in 2020 due to the COVID 19 pandemic, as its activities were suspended

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or, to a lesser extent, limited to deliveries by delivery [3]. However, concerning Peru, this sector, in which food concessionary companies are found, did not have a very different scenario; by March 2020, the growth of this sector fell by 42.35% compared to March 2019, the first drop after 35 months of continuous growth [2].

In this sector, it is essential to efficiently manage the supply chain, meaning all processes ranging from negotiation with suppliers to delivering the product or service to the customer. Because if this were to break down, it would trigger a series of problems with the possibility of not giving a good customer service, such as untimely delivery of orders [4].

On the other hand, in this case, the study of an SME in the food sector identified that the main problem might be a flawed supply system due to poor inventory management and control because of an incorrect application of systematic control methods. These erroneous practices are closely related to the characteristics of the raw materials since they are perishable, causing possible additional costs by increasing the level of raw material wastage. In addition to this, the variable demand becomes an obstacle for the company to increase its response capacity.

This context is persistent in companies in the food sector, affecting their competitiveness and efficiency to meet delivery times. This research aims to include good practices in managing storage and purchase planning of raw materials to solve the described problem. Thus, to achieve good practices, a model based on Lean Logistics and Lean Warehousing methodologies was developed to identify processes that do not add value, reduce logistics costs and delivery times. This improvement model combines the tools of ABC multi-criteria, 5S, BPM, FEFO, and Forecasting.

A literature review of the actual databases found little information on the working models of Lean Logistic and Lean Warehousing for SMEs in the food sector, especially in Latin America, since most of the research focuses on large companies with high purchasing power. Therefore, this literature review will be developed in the state-of-the-art section, where some success cases will also be exposed to give bibliographical support to the research. For this reason, the need for this research arises. Having said the above, this article presents the analysis of the identified problem of the case study in the explanatory contribution of the qualitative and quantitative analysis of the causes and the proposed model as a possible solution.

2 STATE OF THE ART

2.1 Order fulfillment in the industry

It is considered a competitive advantage to meet delivery deadlines on time because nowadays, companies are increasingly seeking to

provide better customer service by generating more excellent value [5] - [6]

Through a literature review, the use of the MRP tool was used by the authors [7]–[9] to have a supply plan of raw materials at the exact time and the right quantities, through planned purchases avoiding delays in the production area thus improving its responsiveness, as well as contribute to the generation of value. [7], in addition to meeting the changing needs of customers.

The author [9], who implemented the tool, achieved a 56% reduction in the cost of materials since there were no more cost overruns due to last-minute purchases or excess in the quantities purchased. Furthermore, as a result of the use of the MRP tool by the authors [10] was obtained result the reduction of stock breakage from 7% to 1%; this tool was also used in the model of [7], resulting in an increase in customer satisfaction by 12% and an increase in the duration of inventories from 62% to 84%.

On the other hand, the authors [11] emphasize increasing productivity and reducing costs by overcoming the barriers to implementing various methods and tools, most closely related to the human factor and the nature of perishable inputs. Furthermore, it emphasizes the absence of awareness to identify the causes of the aspects to be improved.

Finally, the authors [12] investigated implementing Lean and Six Sigma tools in the food industry. They corroborated that it was possible to increase productivity and reduce costs by overcoming the barriers to implementing various methods and tools, most closely related to the human factor and the nature of perishable inputs. On the other hand, it emphasizes the absence of awareness to identify the causes of the aspects to be improved.

2.2 Lean Warehousing - Lean Logistic

The paper [13] through a survey conducted, which shows lean as little relevant in improving external relations with customers and suppliers, so that lean tools are very little considered in the management of the supply chain and even more so in SMEs, which often lack knowledge of the methodology above, application, and benefits that would entail the same.

One of the lean tools found in the Lean warehousing methodology is the multi-criteria ABC tool; according to their article [14], [15] is a well-established categorization technique based on Pareto Principle to determine which items should be prioritized based on various criteria in inventory management. Finally, the case of the article by [16], where after using such a tool, the lead time is reduced by 5.7 minutes.

On the other hand, the inventory control tool FEFO (First Expired, First Out) aims to reduce the number of losses of perishable supplies to avoid non-conformities in the warehouse area. The case study of [17] observed that after having implemented it, the number of expired supplies was reduced by 100%; In addition, the delay in the shipping process was reduced by up to 19%.

Finally, Japan's 5S technique (seiri- classification, seiton- organization, seiso-cleanliness, seiketsu-hygiene, and shitsuke-discipline) seeks to improve and maintain optimal conditions for the organization. Since working in a clean and tidy environment promotes workers' safety and health and offers a safe product or service. Furthermore, this technique seeks workers' participation, assuming

their responsibility based on good practices based on orderly work [18].

Through a literature review of case studies, the [19], [20] applied the 5s tool to allow standardizing, ordering the warehouse area, thus facilitating the management of materials and achieving an efficiency between an approximate range of 58% and 67% percentages.

2.3 Lean Warehousing/ lean logistics in the food sector.

Few authors have decided to direct their research towards a solution that uses models with lean tools that are adapted according to the demands of the food industry and specifically SMEs, despite being an excellent opportunity to make them more competitive with each other. It affirms it [21], highlighting the improvement of quality, the reduction of lead time and waste; It also highlights the importance of understanding the lean method by the participants and the constant evaluation of the changes made [22]. It was identified according to the articles that the multi-criteria ABC classification is implemented to detect the raw materials that are used in greater quantity, as well as those that require greater attention in terms of care, considering that it must be adjusted to the characteristics of the product, the environment and control raw materials efficiently [23].

The FEFO tool helps to identify the level of the valuable life of raw materials and seeks greater logistical efficiency and cost reduction through the proper use of them, ensuring the quality of these and the product or service offered [17], [24], [25]. According to [26], FEFO and 5s tools state that for food and beverage companies, the adjustment of production and stocks is a challenge due to the perishable characteristic of the latter, so it is crucial to identify a model that fits the context of food companies to achieve better costs. The author [26] used the simple exponential smoothing forecasting method to perform a medium-term planning managing to reduce the planning error by 5%.

On the other hand, according to [26] in food companies "perishable inputs," the adjustment of production is a challenge, so it used the forecasting method simple exponential smoothing, because of the previous forecast added by the error and corrected using a smoothed coefficient, reducing the error by 5%.

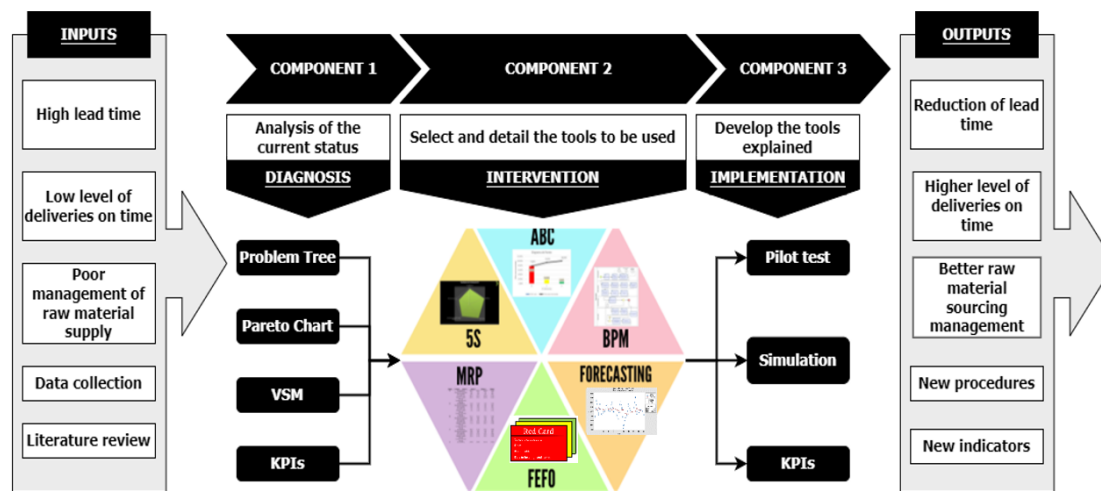
Finally, to [9], [20] BPM (Business process management) is a discipline focused on process management. Besides that, it has many focuses: product improvement, organizational culture, customer satisfaction, information analysis, and process efficiency. By applying the BPM developed by [20], the result was an increase in process efficiency of 15.47%.

3 PROPOSED MODEL

To realize the proposed model, the multi-criteria ABC tool will be implemented in the warehouse area to classify the raw materials and thus improve the distribution and storage based on criteria with its importance of use and prolongation of its useful life. Next, the 5S tool will improve order, cleanliness, and work standardization in the warehouse area. Subsequently, the FEFO tool will be put into practice to control the useful life of the raw materials to ensure that they are used before their expiration date. On the other hand, the MRP tool will be implemented to plan raw material purchases better. In addition, the implementation of GMP is proposed to improve the

Table 1: Comparison table of the proposal components vs. State of the Art

Author	References	Loss of perishable products due to spoilage	Inadequate demand Projection	Inadequate supply planning and control of raw materials
Castro Zuluaga et al., (2011)	[27]	ABC Multicriteria		
Barbosa et al., 2015)	[26]		Forecasting	
Espinoza-Camino et al., (2020)	[17]	FEFO, 5S		
Del-Castillo-Monroy et al., (2020)	[7]			MRP
Retamozo-Falcon et al., (2019)	[20]		BPM	
Proposal		ABC multicriteria, FEFO, 5s	Forecasting, BPM	MRP

**Figure 1: General view of the model Lean Warehousing –Lean**

raw materials purchasing policies. Finally, a forecasting method, simple exponential smoothing, will be developed to have a demand forecast closer to reality.

3.1 Model components

3.1.1 Component 1: Problem identification. Information will be collected through interviews and audits of the company under study to identify the problem. Afterward, the most crucial problem will be captured in an Ishikawa diagram to develop the possible causes, then these causes will be quantified in a Pareto diagram. Then, the results will be captured in a Problem Tree to achieve a better understanding through a visual map. Finally, a VSM (Value Stream Mapping) will be developed, which will allow a panoramic view of the times of the current situation.

3.1.2 Component 2: Intervention. With the above information, the classification criteria of the raw materials of the ABC multi-criteria tool will be established to relocate and optimize the warehouse

area. On the other hand, instruments such as red, green, and yellow traffic light cards will be used, each color representing a range of useful lifetimes. These cards will allow the deployment of the FEFO tool. Next, for implementing the 5S tool where there is more disorder and loss of material in the warehouse of supplies, a macro will be developed that will allow us to supervise the improvements.

On the other hand, it is proposed to implement a master production plan using the MRP tool, Material Requirement Plan. In addition, it will be proposed the implementation of technical datasheets of the raw materials required for each dish offered by the company. Next, we will use the simple exponential smoothing method, Forecasting, using Minitab 19 1.1.0 software, to determine a demand more in line with reality. The development of these two tools will help reduce extra purchases and the lost time in the same ones.

Finally, the BPM tool will describe the current business process. Then, the process will be divided into sub-processes to schematize the flow of the sub-processes, facilitating the understanding of the

Table 2: Table of indicators

Indicator	Formula	Use
On-time delivery – OTD (%)	$\frac{\text{Orders delivered on time}}{\text{total orders delivered}} \times 100$	Shows the percentage that presents the orders received on time concerning the number of orders delivered in total.
5S Audit	$\sum$ The score resulting from the macro	It quantifies the degree of compliance of each 5S component by adding up the results obtained by each of them.
Amount of Expired Stock of Supplies	$\sum$ Expired raw materials	It quantifies the total amount of expired raw materials in PEN; the objective is to corroborate whether there is a priority for the use and care of the oldest raw materials of the type A raw materials resulting from the ABC multi-criteria analysis.
Average Mean Absolute Error -MAPE (%)	$\frac{\sum(\text{Real Demand} - \text{Forecast Demand} / \text{Real Demand}) \times 100\%}{n}$	Determines the percentage difference between the actual demand and the p resulting from the simple exponential smoothing method for a given month, considering that "n" is the sample size.
Time Set-Up (%)	$\frac{\text{IST} - \text{FST}}{\text{IST}} \times 100$	It allows calculating the percentage of improvement concerning the time of the initial situation (IST) and the total time of the final situation (FST) over the IST. This indicator will be used to measure the improvement of the Lead time of the company's service.
Number of extra purchases	# Number of times additional purchases	This indicator details the number of times additional purchases are made in the workweek.

functioning of the business process, analyzing it, and proposing improvements. Finally, the improvements will be documented, detailing how to carry out the activities, processes, and functions of the personnel belonging to the commercial area.

3.1.3 Component 3: Implementation. The development of the "ABC" tool consists of redesigning the materials warehouse concerning the criteria for extending its useful life. For this reason, to formalize its compliance with placing the raw materials in their respective established place, an inventory storage policy will be drafted. Furthermore, for implementing the FEFO tool, an Excel spreadsheet will be used that contains three sheets related using a macro. The function of the first sheet is to record the quantity and date of raw materials purchased, and the second is to record the number of raw materials to be used in the kitchen.

Finally, the third sheet will be updated each time the macro is run, adding and decreasing the quantities and dates inserted in sheet one and sheet 2 to the previous existing quantities. Traffic light cards will be used with the quantities, the red ones for products whose shelf life is less than two weeks, while the yellow ones between 2 to 4 weeks and the green ones in a range of 5 to 7 weeks. This

signaling system should be updated weekly through a record of the movement of the warehouse that will be a support for decision making in the purchase of raw materials. Next, the 5S tool will be implemented through a monitoring macro divided into five tabs, in which checklists will be found and evaluated between a range of 1 to 4, being one bad, two regulars, three good, four, and five very good. Each element that makes up the tool will be averaged together, and this tab will be done by the warehouse manager each week.

Then, the development of the resulting purchase orders and the data allows executing a work order of the MRP model material plan in Excel macros. On the other hand, for implementing the simple exponential smoothing model, first, the past demand values for the last 12 months will be collected in Minitab 19 1.1.0 using the smoothing method.

The indicators presented in Table 1 are important since these will allow measuring the performance of the current situation and the situation after implementing the proposed model; besides being of support to analyze the compliance of the proposed model, on the other hand, it is considered that the leading indicator is the KPI of On-time delivery (OTD).

4 VALIDATION

4.1 Initial diagnosis

In analyzing the company's current situation, the main problem detected is the high rate of late deliveries, which represents 31.11% of the total orders delivered per year, being compared with the SMEs of the same sector presents on average 10% of its deliveries to the customer are late [28]. Currently, the consequence of this problem has an economic impact of 113,380.00 PEN per year, representing 7.74% of total costs. It is caused by several factors such as extra purchases, discounts for repeated delays according to policies managed with the customer, losses due to deterioration of raw material, and mobility expenses to selected locations where only raw material can be supplied. The leading causes of the problem are a) loss of perishable products due to spoilage (47%), b) inadequate demand projection (35%), c) insufficient planning and control of raw materials (18%). The results of the proposed model development will be shown below, in addition to the indicators that will allow a complete and correct evaluation.

4.2 Validation design and comparison with initial diagnosis

To validate this research, a pilot test was conducted for three months, covering the months of May, June, and July 2021, with a sample of services provided to a particular client.

On the other hand, the first tool implemented was the ABC multi-criteria, where the results showed that zones A, B, and C represent 76.85%, 16.32%, and 6.83% of the raw materials, respectively.

To complement the previous tool, the FEFO tool was implemented in which it was found that the total amount of deteriorated raw materials belonging to zone A was 20 kilos per week, for this reason as an expectation is intended to reduce by 50%.

As for the prediction of simple exponential smoothing demand in Minitab 19, 1.1.0 was obtained; as a result, the MAPE (Mean Average Absolute Error) was 4.1%. On the other hand, the 5S tool at the end of its application and compared to the result of the initial situation, an increase of good practices of 27.27% was observed.

Next, the BPM tool was implemented, which because of the analysis of the purchasing process was 63%. Finally, the MRP tool was used, which resulted in a 20% reduction in stock.

4.3 Simulation of the proposal

A prospective validation technique was developed using Arena 16.1 software to validate the proposal. A simulation of the current situation and the development of the improved proposal was designed with a confidence level of 95%. The improvement proposal seeks to reduce the delivery time, so it was decided to simulate placing orders for one week. The data provided by the company that was used in the simulation were from June to August 2021. For this reason, we used the input analyzer to determine the distribution of each type of data. Picture 2 shows the simulation of the current situation, which fits perfectly to reality, in addition to describing the process of the extra purchase. Image 3 shows the simulation of the process integrating the proposed model. On the other hand, the optimal number of replications to be simulated in Arena 16.1 was calculated by the VA time (average) data of the output analysis

of 15 runs. Finally, the optimal run size of the current system was 29.74, which was rounded to 30, and the improved system was 29.57 runs which were also rounded to 30.

The following image illustrates the simulation of the current and improved situation

It should be emphasized that for the realization of this research, previous research was considered to formulate the integrated model, which will be detailed in the following table. However, most of these are not studies focused on SMEs in the food sector. In addition, they do not implement more than two tools at a time of the five proposed in this research. Nevertheless, it was possible to evaluate each proposed tool and technique despite this.

5 DISCUSSION

The following table illustrates the results of the indicators in Table 1 of the current and improved situation of the simulation.

Where:

- O1 On-time delivery OTD
- O2 5S Audit
- O3 Amount of Expired Stock of Supplies
- O4 Average Mean Absolute Error
- O5 time of the system
- O6 number of extra purchases
- E0 Current Situation
- E1 Improved Situation

The following are the results obtained from the implementation of the proposed model in which after having trained workers on what each tool consists of, its importance, how it will be carried out, emphasizing on the KPI that will measure its performance, a pilot test was conducted in one week of the 5S, FEFO and ABC tools since it was considered less challenging to simulate it. This implementation allowed optimization of the storage space, as well as the correct order and good location of raw materials; that is, the management of raw materials in the warehouse was improved and was reflected in the corresponding indicators of each of the tools with an increase of 37.2% and 12.46% of the 5S tools and ABC and FEFO, respectively. Furthermore, once the pilot test and the simulation were completed and through the simulation, the results obtained showed an improvement such as the reduction of extra purchases by 50% after using the MRP tool, but not before applying a good forecast, reducing its error by 9.5%, concerning the error before applying the tool. Thus, after applying all the tools, the lead time of the process was reduced by 11%, achieving the desired result.

6 CONCLUSION

It is concluded that the improvement of the supply management can reduce the lead time of the order elaboration process, thus fulfilling the KPI on-time deliveries. On the other hand, it was concluded that the importance of the understanding and involvement of the workers is crucial during the process since it is closely related to the results, likewise. Furthermore, to sustain the inserted changes, it is necessary to perform audits to ensure the correct implementation of each tool. Finally, it can be concluded that the simulation as an option that exemplifies the whole process involved in the case study and allows to obtain a model of the different contexts that

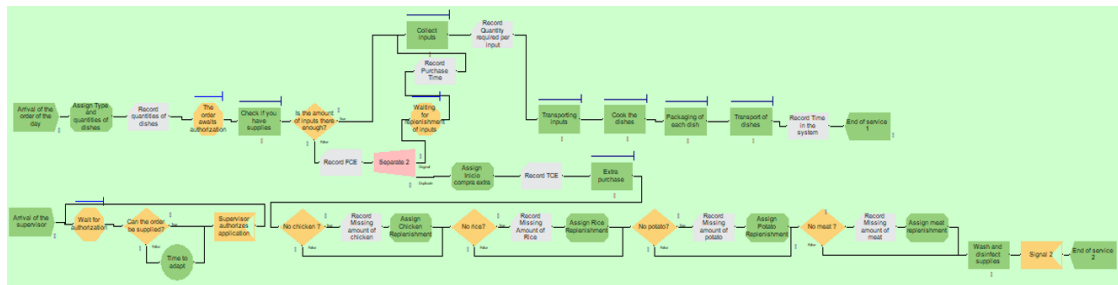


Figure 2: Current Situation

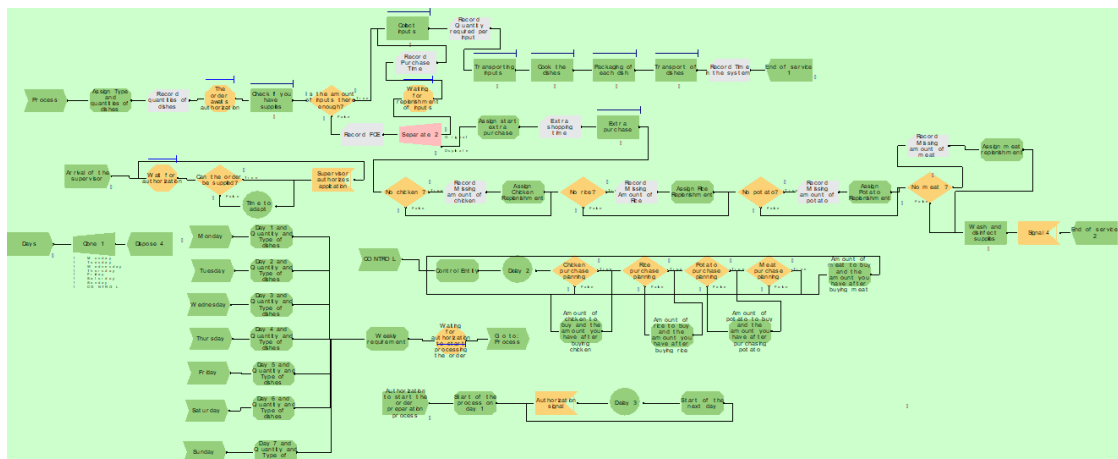


Figure 3: Improved Situation

Table 3: System representation

Objectives						
Situation	O1	O2	O3	O4	O5	O6
E0	21.43%	27	13	2.1	301.45	6
E1	64.29%	43	7	1.9	279.63	3
Variation	42.8%	37.2%	12.46%	9.52%	7.2%	50%

could arise in addition to obtaining the results of the same, without running risks such as generating losses to the company, in addition to obtaining it in less time.

It is concluded that after the pilot test of the FEFO, ABC, and 5S tools, as well as the subsequent simulation of the process considering the other tools such as Forecasting, MRP, BPM of the small food concessionaire company, a significant improvement was obtained in time set up of the process as well as an increase in the number of orders delivered on time, by 7.2% and 42.8% respectively. On the other hand, some of the causes affecting both Time Set Up and on-time delivery of orders were extra purchases, which were reduced by 50%, and the demand forecast error by 9.5%.

The results were obtained thanks to the integration of all the tools. On the one hand, there is forecasting to improve the demand forecast linked to the use of the MRP tool, which allowed to have a better planning of the raw materials required for the preparation

of the order and all this is also associated with the improvement of trade policies with suppliers through a standardized process all these tools contributed to the reduction of waste and extra purchase of raw materials.

On the other hand, the 5S, ABC, and FEFO tools contributed to the company's better management of raw materials in the warehouse, thus ensuring the safety of all raw materials. However, the main objective of using these tools is to ensure the lifetime of the raw materials to generate extra purchases, which would mean extra cost and loss of time for the company.

It can be concluded that the integrated model proposed in the present research managed to improve the main objective; that is to say, the lead time of the process was improved by 11%, thus demonstrating an improvement opportunity for the company to become more competitive. However, to implement the present model, it is necessary to consider the company's size; on the other hand, it

was observed that although the model indeed managed to improve the current situation, some indicators did not reflect the expected improvement.

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